Lecture 08: Equity Valuation: FCFE

Questions

Choose one answer. Keep full precision until the final result. Worked solutions follow the complete question set.

1. From operating income to equity cash flow

All amounts are in $ millions. EBIT is 250, interest expense is 30, and the tax rate is 25%. $D\&A$ is 25, CapEx is 70, operating working capital increases by 20, and net debt issuance is 15. In the simplified income statement used in class, what is current FCFE?

A. $137.5 million
B. $100 million
C. $165 million
D. $115 million

2. Use the leverage definition from class

Under the simplified balance sheet $A = D + E$, assets are $500 million and financial debt is $150 million. The firm maintains $L = \frac{D}{A}$ and finances new investment in that proportion. Net CapEx is $60 million and ΔWC is $20 million. What is equity reinvestment ER?

A. $56 million
B. $24 million
C. $45.71 million
D. $61.54 million

3. Reinvestment generates sustainable growth

A firm has current NI of $180 million, beginning book equity of $900 million, and equity reinvestment of $54 million. Assuming a constant reinvestment rate and ROE, what sustainable growth rate g should be used?

A. 20.0%
B. 30.0%
C. 6.0%
D. 9.0%

4. Do the full FCFE valuation

All amounts are in $ millions. NI is 200, Net CapEx is 50, ΔWC is 30, and $L = \frac{D}{A} = 25\%$, maintained for new investment. ROE is 10% and k is 12%. Using stable growth $g = \frac{ER}{NI} \times ROE$, what is total equity value?

A. $1,602.22 million
B. $1,166.67 million
C. $1,555.56 million
D. $1,373.33 million

5. Compare DDM and FCFE on the same basis

A DDM produces a value of $42 per share. There are 20 million shares. An FCFE model produces total equity value of $1,050 million. By what percentage is the FCFE value above the DDM value, using DDM as the base?

A. 2,400.0%
B. -20.0%
C. 20.0%
D. 25.0%

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Worked solutions

1. From operating income to equity cash flow

Answer D. $115 million

Step 1. Calculate earnings after interest and taxes.

We want cash available to shareholders, so start with earnings after interest and taxes.
EBIT is still before the lenders' interest payment.
Deduct interest first, then apply the tax rate to the remaining income.

$$NI = (EBIT - \text{Interest})(1 - t)$$

$$= (250 - 30) \times 0.75 = 165 \text{ million}$$

Step 2. Find reinvestment financed by equity.

Next, work out how much investment shareholders must finance.
Subtract depreciation from capital expenditures, add the working-capital increase, and deduct net new debt.
Borrowing supplies part of the cash needed for investment.

$$\text{Net CapEx} = 70 - 25 = 45$$

$$ER = \text{Net CapEx} + \Delta WC - \Delta D = 45 + 20 - 15 = 50$$

Step 3. Subtract equity reinvestment from net income.

What remains after equity reinvestment is FCFE.
I also show the expanded cash-flow calculation below so you can check the signs.
Depreciation is added back; capital spending and working-capital investment use cash.

$$FCFE = NI - ER = 165 - 50 = \boxed{115 \text{ million}}$$

$$FCFE = 165 + 25 - 70 - 20 + 15 = 115$$

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2. Use the leverage definition from class

Answer A. $56 million

Step 1. Calculate the debt share of total financing.

Here our leverage measure is debt divided by assets.
With the simplified balance sheet used in this question, assets are financed by debt and equity.
Therefore, one minus this debt share is the equity share.

$$L = \frac{D}{A} = \frac{150}{500} = 0.30, \qquad 1 - L = 0.70$$

Step 2. Add the investment that needs financing.

Add the two investment needs before splitting the financing.
Under the stable-leverage assumption, the firm borrows enough to cover 30% of this new investment.
This is an assumption about financing, not an accounting identity that holds for every firm.

$$\text{Net CapEx} + \Delta WC = 60 + 20 = 80$$

$$\Delta D = L(\text{Net CapEx} + \Delta WC) = 0.30 \times 80 = 24$$

Step 3. Allocate the rest to equity.

Shareholders finance the remaining 70%.
We can calculate that directly or subtract the implied borrowing from total investment.
Both routes should give the same equity reinvestment.

$$ER = (1 - L)(\text{Net CapEx} + \Delta WC)$$

$$= 0.70 \times 80 = 80 - 24 = \boxed{56 \text{ million}}$$

3. Reinvestment generates sustainable growth

Answer C. 6.0%

Step 1. Measure the share of earnings reinvested.

First ask how much of income goes back into the business on behalf of shareholders.
Divide equity reinvestment by net income.
That gives us a reinvestment rate, not a growth rate yet.

$$ERR = \frac{ER}{NI} = \frac{54}{180} = 0.30 = 30\%$$

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Step 2. Measure the return on beginning equity.

Then ask how productive that equity is.
Use net income relative to beginning book equity, as specified in the question.
ROE measures the return on equity; it does not tell us what fraction of income is reinvested.

$$ROE = \frac{NI}{E_{\text{beginning}}} = \frac{180}{900} = 0.20 = 20\%$$

Step 3. Combine the amount reinvested and its return.

Multiply the reinvestment rate by ROE.
The intuition is straightforward: growth depends on how much we reinvest and what return we earn on it.
This assumes those rates remain sustainable.

$$g = ERR \times ROE = 0.30 \times 0.20 = \boxed{6\%}$$

4. Do the full FCFE valuation

Answer A. $1,602.22 million

Step 1. Find equity reinvestment and current cash flow.

I would solve this in a fixed sequence.
First find total investment and the share financed by equity.
Subtract that equity reinvestment from net income to obtain current FCFE.

$$ER = (1 - 0.25)(50 + 30) = 60$$

$$FCFE_0 = NI - ER = 200 - 60 = 140$$

Step 2. Build growth from reinvestment.

Next, calculate growth from the reinvestment rate and ROE.
Do not use the reinvestment rate itself as growth.
Once growth is known, move current FCFE forward one year because the valuation starts with next year's cash flow.

$$ERR = \frac{60}{200} = 30\%, \qquad g = 0.30 \times 0.10 = 3\%$$

$$FCFE_1 = 140 \times 1.03 = 144.20$$

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Step 3. Discount the growing equity cash flow.

Finally, apply the growing-perpetuity formula using the cost of equity.
Check that the required return exceeds growth.
Since our cash flow is in millions for the whole firm, the answer is total equity value in millions.

$$V_{E,0} = \frac{144.20}{0.12 - 0.03} = \boxed{1{,}602.22 \text{ million}}$$

5. Compare DDM and FCFE on the same basis

Answer D. 25.0%

Step 1. Put both estimates in total-equity dollars.

Before comparing the models, check the units.
DDM gives a price per share here, but FCFE gives a total equity value.
Multiply the DDM price by shares so that both estimates describe the same equity claim.

$$V_E^{\text{DDM}} = P_0^{\text{DDM}} \times N = 42 \times 20 = 840 \text{ million}$$

Step 2. Compare values using the requested denominator.

The question asks how much FCFE exceeds DDM, so DDM is our reference value.
Divide the difference by the DDM estimate.
Using the FCFE estimate in the denominator would answer a different question.

$$\text{Premium} = \frac{1{,}050 - 840}{840} \times 100 = \boxed{25\%}$$

Step 3. Check the answer on a per-share basis.

We can verify the percentage by converting FCFE into a price per share instead.
Both approaches must agree.
After doing the arithmetic, I would ask which payout and reinvestment assumptions are more convincing, rather than automatically picking the higher number.

$$P_0^{\text{FCFE}} = \frac{1{,}050}{20} = \$52.50$$

$$\left(\frac{52.50}{42} - 1\right) \times 100 = 25\%$$